

ClearWrite

An Android writing editor that checks clarity locally, explains specific problems, and offers selective AI rewriting while keeping every change under the user's control.

Check locally.
Explain clearly.
Rewrite selectively.

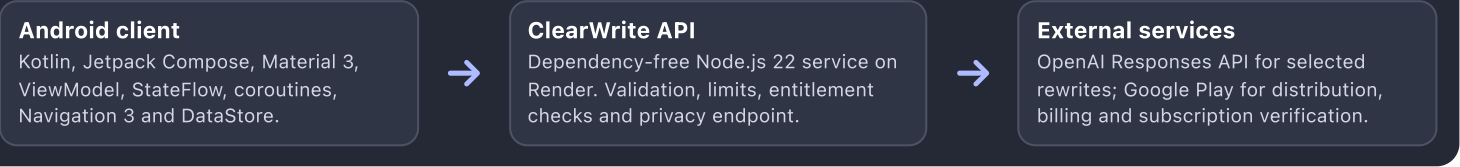
What it solves

Writing tools often flag problems without context or rewrite too much at once. ClearWrite takes users directly to the affected text, explains the concern, and lets them edit manually or request help for only the selected passage. AI supports the writer; it does not replace their voice.

Core experience

1. Type, paste, or import a TXT/DOCX draft.
2. Receive an automatic local clarity score and issue list.
3. Tap a category to highlight and navigate exact passages.
4. Edit manually or request a Pro AI rewrite for any selection.
5. Review, edit, apply, dismiss, or undo every suggestion.

Architecture at a glance



Android app & local analysis

MainScreenViewModel owns the document and UI state. Coroutines debounce analysis and handle persistence, billing, and network work. DataStore keeps drafts, titles, documents, and preferences in private on-device storage.

The deterministic analyzer runs offline and checks sentence/paragraph length, readability, possible passive voice, weakening and wordy language, filler-heavy sentences, repeated openings, and repeated words. ClearWrite Score v2 combines clarity, concision, audience readability, and flow.

10	1,500	10,000
local checks	free words	Pro words

Backend, hosting & AI

The production API runs on Render; its URL is injected at build time. Secrets stay in the hosting environment and never ship in the APK.

An AI request contains only the selection, limited nearby context, requested action/tone, locale, and technical identifiers. The backend validates the request and Pro access, then calls the OpenAI Responses API with Structured Outputs. Instructions require preservation of meaning, facts, names, numbers, claims, and language while returning one or two alternatives.

Requests use store: false. ClearWrite has no document-content database and does not intentionally log drafts or suggestions. Development uses a deterministic mock provider. Production applies timeouts, security headers, a 16 KB body limit, and an in-memory 20-requests-per-minute client limit.

Payments & Pro access

Google Play Billing 9.1 supplies plans and localized prices. After purchase or restore, the backend verifies the token through the Android Publisher API and acknowledges new purchases. AI endpoints recheck entitlement server-side, preventing local-state bypasses. Verified access can be cached for up to 24 hours; an expiring server-issued token supports Play reviewers.

Current scope

The local engine is a clarity checker—not a comprehensive grammar or spell checker. English is the initial focus, and text can score 100 when it matches none of the supported patterns. AI requires Pro access and a network connection.

Privacy model

- Drafts, imported content, titles, settings, and documents stay on-device; there is no account or cloud sync.
- Standard analysis is local and works offline.
- Text leaves the device only after an explicit AI request.
- Google Play handles payment details; ClearWrite processes tokens and entitlement status.
- No third-party ads, sale of drafts, or targeted advertising; production traffic uses HTTPS.

Build, testing & delivery

Gradle builds signed Android App Bundles for Google Play. Unit and Compose tests cover analysis, scoring, offsets, limits, ViewModel behavior, editor flows, imports, issue navigation, and AI suggestions. Node tests cover validation, AI responses, billing gates, reviewer access, health, and privacy.